# Supporting Information: Supporting Information: When a Learned $\sigma$-Profile Is Not One: Diagnosing the Intermediate of a Physics-Embedded Solubility Model

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This document supports the article in five parts, in the order its Supporting Information section lists them: the architecture and the training schedule (§1); the data, the splits and the auxiliary supervision streams (§2); the reading rule declared for each measurement before it was made, with its outcome (§3); what the observations leave undetermined, which is the premise the measurements rest on (§4); and a second system in which a linear free-energy relation stands in for the closure (§5). Every number here is produced by a named script in the deposited repository from a named file in it; the mapping is `MANIFEST.md` in that repository.

## 1 Architecture and training

**The model.** A shared message-passing encoder embeds the solute and the solvent graphs, with a two-layer residual role-specific branch. A set2set readout pools each graph, the two pass through a cross-attention block, and the pair representation $[\, g_s,\, g_v,\, g_s \odot g_v,\, |g_s - g_v| \,]$ feeds both heads. Table S1 gives the widths, the depths, the closure constants and the training hyperparameters; it is the schedule the five checkpoints of record were trained under.

The crystal head predicts $T_m = 100 + 600\,\mathrm{logistic}(\cdot) \in [100, 700]\,\mathrm{K}$ and $\Delta H_{\mathrm{fus}} = S_H\,\mathrm{softplus}(\cdot)$, giving the ideal term $\Phi(T)$ of equation (1); the group-contribution $T_m$ prior is affine-calibrated on the training split before entering the head. The $\sigma$-profile head is a feed-forward network on the same pair representation, ending in a normalised non-negative vector of 51 bins.

The repository carries two further encoder backbones, a graph transformer and a variant that splits messages into dispersive, polar and hydrogen-bonding channels. Neither is used by any run reported here, and no accuracy comparison between backbones is claimed.

**The closure, and what it omits.** The differentiable COSMO-SAC layer uses the standard VT-2005 / Lin–Sandler constants, with the Staverman–Guggenheim normalisers and the profile grid as listed in Table S1.

The layer runs the residual, segment-activity part of COSMO-SAC alone. The Staverman–Guggenheim combinatorial term, the contribution molecular size and shape make to the activ-

Table S1: Architecture, closure constants and training hyperparameters of the five checkpoints of record, and the four hyperparameters on which the direct control was tuned separately; values not listed for the control are shared with the physics arm. Units: $a_{\mathrm{eff}}$ in Å$^2$, the misfit constant in kcal Å$^4$ mol$^{-1}$ e$^{-2}$, the bonding cutoff in e Å$^{-2}$, the normalisers in Å$^3$ and Å$^2$; the profile grid spans $\pm 0.025$ e Å$^{-2}$.

| quantity | value | quantity | value |
|---|---|---|---|
| *encoder* | | *fixed points* | |
| hidden width | 64 | damping $\lambda$ | 0.7 |
| message-passing layers | 3 | segment activity, train / eval | 16 / 30 |
| node / edge feature dimension | 35 / 8 | phase equilibrium, train / eval | 5 / 20 |
| set2set processing steps | 3 | *schedule* | |
| cross-attention layers | 3 | epochs, phases 1–3 | 30 / 70 / 10 |
| pair representation dimension | 512 | learning rate, phase 1 | $2.6 \times 10^{-4}$ |
| dropout | 0.012 | learning rate, phase 2 | $8.5 \times 10^{-5}$ |
| trainable parameters | 5.40 M | learning rate, phase 3 | $8.5 \times 10^{-6}$ |
| *closure* | | optimiser | Adam |
| segment area $a_{\mathrm{eff}}$ | 7.5 | weight decay | $5 \times 10^{-4}$ |
| misfit constant $\alpha'$ | 16466.72 | gradient clipping | 2.0 |
| bonding coefficient $c_{\mathrm{hb}}$ | 85580 | batch size | 64 |
| bonding cutoff $\sigma_{\mathrm{hb}}$ | 0.0084 | *direct control* | |
| coordination number $z$ | 10 | dropout | 0.174 |
| normalisers $r_0$ / $q_0$ | 66.69 / 79.53 | phase-2 learning rate | $8.6 \times 10^{-4}$ |
| profile grid, bins | 51 | weight decay | $4.2 \times 10^{-5}$ |
| | | gradient clipping | 2.54 |
| | | epochs | 110 |

ity coefficient, is omitted, because it is a function of cavity volume and area rather than of the profile and so carries no information about the quantity under study. A substitution that replaces the profile and leaves the combinatorial term in place would move both terms and the attribution would not be clean.

Restoring the term would change some of the article's magnitudes and none of its signs. It vanishes for a pair of a molecule with itself, so the reference-state violation of Section 3.4 is unaffected. It would change the magnitudes in Sections 3.5 and 3.7, where the size of the effect is the result. It needs molar volumes, which the source database carries but the extracted table used here does not, and which the network could supply with a second scalar head beside the one that already predicts cavity area; its omission is a choice made when the table was extracted rather than a constraint.

**Two fixed points, and the iteration counts they are run at.** The pipeline contains two nested fixed points, each with its own iteration count, and the counts differ between training and evaluation.

The two counts and the damping are in Table S1. The phase-equilibrium fixed point is the damped map $x_2^{(k+1)} = \lambda \exp(-\Phi(T) - \ln \gamma_2(x_2^{(k)}, T)) + (1-\lambda) x_2^{(k)}$, run at the same $\lambda$ as the segment loop, which is one configuration field read by both.

A count is a free parameter between fitting and scoring, so a checkpoint scored at a count far from the one it was fitted at is evaluated through an operator it was never fitted against. For the checkpoints reported here the exposure is bounded directly. Table S2 sweeps the segment count with weights and rows held fixed: converging the solver makes the learned arm slightly *worse*, over an interval 0.021 wide, and moves the contrast between the two arms by $-0.008$ between the evaluation count and the fixed point, 1.9% of itself. That movement is not a direction: three of five seeds carry its sign, the per-seed range $[-0.031, +0.002]$ spans zero, and one seed carries almost all of it.

The aggregate is insensitive because the per-

Table S2: Seed-mean $\ln x_2$ mean absolute error against the number of segment-activity iterations, with weights and rows held fixed. The training count is 16 and the evaluation count 30; the fixed point is reached by about 300. Contrast is the substituted arm minus the learned one.

| segment iterations | learned | substituted | contrast |
|---|---|---|---|
| 8 | 1.9155 | 2.2300 | +0.3145 |
| 16 | 1.9195 | 2.3061 | +0.3866 |
| 30 | 1.9275 | 2.3356 | +0.4081 |
| 100 | 1.9362 | 2.3370 | +0.4008 |
| 300 | 1.9364 | 2.3368 | +0.4004 |

row shifts cancel, not because they are small. At 30 iterations the median row moves $5.9 \times 10^{-4}$ against the converged solve, its ninety-fifth percentile 0.136 and its worst row 2.03 in $\ln x_2$. The two profiles also converge at different rates, in the direction the asymmetry predicts: at 30 iterations the segment residual is a median $9.7 \times 10^{-4}$ with a ninety-fifth percentile of 0.038 and a worst case of 0.073 in $\ln \gamma_2^{\infty}$ on the tabulation, against $2.4 \times 10^{-3}$, 0.077 and 0.162 on the learned profile, so the broader learned profile is about twice the slower wherever the residual is large. What the count does reach is undercounting: at eight iterations the contrast reads +0.31, understating it by 0.094. The exposure is therefore a head scored *below* the count it was fitted at, which is not the configuration of any run reported here.

**Optimisation and curriculum.** Training is a fixed three-phase curriculum: 30 epochs of auxiliary-property pretraining, 70 of full solid-liquid equilibrium training, and 10 of low-rate stabilisation, at learning rates $2.6 \times 10^{-4}$, $8.5 \times 10^{-5}$ and $8.5 \times 10^{-6}$. The optimiser is Adam with weight decay $5 \times 10^{-4}$, gradient clipping at 2.0 and batch size 64.

The direct control shares the encoder, interaction and readout verbatim and emits $\ln x_2$ directly, with a thermometer encoding of temperature in place of the dependence the physics arm receives through $\Phi(T)$ and the closure. It was tuned separately: dropout 0.174 against 0.012, second-phase learning rate $8.6 \times 10^{-4}$ against $8.5 \times 10^{-5}$, weight decay $4.2 \times 10^{-5}$ against $5 \times 10^{-4}$, gradient clipping 2.54 against 2.0, over 110 epochs equal to the physics arm's total budget at the same width and depth. Width, depth, batch size and epoch budget are matched and the optimisation is not, so any difference between the two arms is a difference between two separately tuned pipelines and is not attributable to the thermodynamic bottleneck alone. A hyperparameter-matched control was not run.

**Objective.** The objective is a weighted sum. Its registry holds 32 terms, of which 17 are hard-wired to zero and two more default to zero through the configuration: disabled experimental scaffolding, released with the code so that a reader of the configuration can see what is off. Table S3 lists the terms carrying a non-zero default weight, together with the $\sigma$-profile earth-mover term, which is applied outside the registry when the grounding stream is on. Per-phase weight overrides are pinned in each run's resolved configuration, which is deposited with the checkpoint.

One entry in that table is a diagnostic rather than a penalty. The monotonicity term differentiates $\ln x_2$ with respect to temperature without retaining a graph, so it enters the reported objective and no gradient: varying only its weight moves the reported total and leaves every parameter gradient unchanged. Its logged value was 0 at every evaluation, and a randomly initialised model scores 0 as well, so that zero records the construction and not the training — $\ln x_2 = -\Phi(T) - \ln \gamma_2$ is non-decreasing in $T$ along the solver path wherever $\Phi$ falls with $T$. Monotonicity in $T$ is carried instead by the same-pair temperature-ranking and local van't Hoff terms, which are zero by registry default and enabled only by the per-phase overrides, so they do not appear among the defaults above.

**The activity-model variants that are not reported.** The code implements two further activity models as controls: a fitted NRTL excess-Gibbs-energy model, and its symmetric one-parameter collapse in which the two interaction parameters are tied to a single pre-

Table S3: Loss terms with a non-zero default weight. The $\sigma$-profile earth-mover term is not a registry entry and is active only while the grounding stream is on. A further 17 registered terms are hard-wired to zero. The monotonicity entry carries a weight but no gradient, and is listed as a diagnostic for that reason.

| role | term | default wt. |
|---|---|---|
| task | solubility (bin-weighted Huber on $\ln x_2$) | 1.0 |
| crystal grounding | melting point $T_m$ | 0.3 |
| | fusion enthalpy $\Delta H_{\mathrm{fus}}$ | 0.3 |
| activity grounding | infinite-dilution $\ln \gamma_2^{\infty}$ | 0.5 |
| | finite-composition $\ln \gamma_2$ | 0.5 |
| | Hansen parameters | 0.2 |
| | $\sigma$-profile earth-mover | stream |
| regulariser | physics preference | 0.1 |
| | adaptive-correction residual | 0.01 |
| | NRTL $\tau$ | 0.01 |
| control coupling | direct-arm negative log-likelihood | 0.2 |
| | direct-arm regulariser | 0.05 |
| architecture | mixture-of-experts balance | 0.02 |
| diagnostic | solubility monotonicity in $T$ | 0.1 |

dicted $\ln \gamma_2^{\infty}$. Neither is used by any measurement in the article, which concerns the $\sigma$-profile and therefore requires the profile-based closure. Their weights were not retained, and no comparison between them and the COSMO-SAC arms is made here: the arms were separately tuned, and a difference between separately tuned pipelines is not a property of the activity model.

# 2 Data, splits and the auxiliary supervision streams

**Sources.** Solubility is BigSolDB 2.0.[S1] Crystal $T_m$ and $\Delta H_{\mathrm{fus}}$ come from an open melting-point pool together with a group-contribution prior; melting-point tables are auto-detected as degrees Celsius or kelvin, because a double conversion of an already-absolute table shifts every $T_m$ by $+273\,\mathrm{K}$ and $T_m$ enters the ideal term directly. Reference $\sigma$-profiles are the VT-2005 compilation[S2] as redistributed with Ref.,[S3] ingested to 1432 compounds together with the COSMO cavity volume. Infinite-dilution activity coefficients are from ThermoML.[S4] The acid-dissociation data of §5 are the OPERA compilation,[S5] 7904 records, from which a uniform net-formal-charge filter leaves 1004 rows amenable to a Hammett treatment, 158 meta or para and 846 ortho or heteroaromatic.

The reference profiles are computed for a single conformer and protonation state. Mismatch between that state and the one a measurement was made in is error in the reference, and it is not signed: it displaces the quantity every departure in the article is measured against. The article therefore compares the learned profile with one tabulation throughout and makes no claim about the tabulation itself.

**Coverage, and why the matched set is small.** The set on which a learned profile and a reference profile can both be scored against a measured activity coefficient is limited by the intersection of the two, not by either alone. At 298 K it holds 85 pairs over 46 solutes and 19 solvents, and matching on RDKit canonical SMILES and on InChIKey return the same set. They did not always. A standard InChI whose mobile-hydrogen layer leaves the tautomer open was being resolved to a minor form, so formamide entered as the imidic acid, N-methylformamide likewise and caprolactam as the lactim, and 27 records then failed to match a profile that was present under the amide form. That is not a tautomer crossed in the matching

but one introduced in our own conversion, since the records' InChI identifies the amide; the conversion is corrected, the two rules agree, and this work therefore reports no trade-off between them. The article uses the canonical-SMILES rule because it is the rule the deployed $\sigma$-oracle uses.

Whether a larger database would enlarge that set is measured here rather than inferred from the solute margin alone. Against PubChemQC COSMO, the largest open $\sigma$-profile database at 334714 distinct molecules, which is 148 times the deposited compound list, joint coverage of the 5608 labelled test rows grows from 437 rows over 11 of the 147 test solutes to 941 rows over 23. A pool 148 times larger buys a matched set 2.2 times larger, and the limit is structural rather than accidental: that database stops below 250 Da and 19 heavy atoms, while the 124 test solutes it does not hold carry a median of 25 heavy atoms against 14 for the 23 it does. The binding side is the solute. Of the labelled rows 4496 fail on the solute alone while 60 of the 70 test solvents are covered, so a database of this kind supplies the solvent channel and leaves the drug-like solutes to be computed.

**The split.** The split is by Bemis–Murcko scaffold:[S6] molecules are grouped by ring system plus linker with side chains stripped, and no scaffold appears in more than one partition. Every test metric in the article is computed on the 5608 labelled test rows, which span 147 solutes.

The seeded split is not stable across versions of the data-preparation code. Regenerating it reshuffles which scaffolds land in the test partition, which orphans every checkpoint and every number measured on the previous split. All numbers in this work are therefore computed on a single pinned split, whose per-file SHA-256 each run manifest records and whose files are deposited with the checkpoints rather than regenerated.

**The auxiliary supervision streams.** An auxiliary stream is a sidecar table in the processed-dataset format whose rows are self-solvent, carry no solubility label, and switch on the mask of one target only. Two streams feed the runs reported here: a crystal stream carrying melting points, and a $\sigma$-profile stream carrying reference profiles. A stream builder applies two guards and reports both in its build summary. The first excludes any pool molecule whose Bemis–Murcko scaffold appears in the validation or test partition, and aborts rather than proceed if a split file is missing. The second reports how many pool molecules are not already labelled for that target in the training partition, by canonical SMILES, because a stream that is mostly duplicates tests nothing and the null it produces would be read as evidence that grounding does not help.

**One build violated the first guard, and the exposure it leaves.** A correct build of the $\sigma$-stream retains 1319 of the 1427 VT-2005 molecules that parse into the pool; the stream file was at one point rebuilt with the exclusion pointed at the splits of a small debug dataset, and that build carries 106 rows whose scaffold is in the real test and validation union, 91 of them held-out solutes by canonical SMILES. What a leak can act on is the labelled test set, and there the exposure is smaller: of the 147 labelled test solutes, 7 have their own reference profile in the defective stream and 8 have a scaffold relative in it (297 and 351 of the 5608 rows, 5.3% and 6.3%), and for a scaffold split that solute coverage rather than the row fraction is the load-bearing statistic. The correctly guarded build leaves none of either, and the crystal stream's build on disk records the correct exclusion, so among the streams feeding these runs the defect is confined to the $\sigma$ one. Per-run stream provenance was not logged, so the deposited artifacts cannot certify which build fed which arm and the above is a bound rather than an attribution; the runner now records the stream hash alongside the split hash.

**Compute and determinism.** The five checkpoints of record were trained on one machine from one commit, at seeds 42 to 46; each manifest records the commit, the configuration, the seed and the environment. Retraining the same arm on a different GPU architecture

does not reproduce a checkpoint to within seed spread, so a comparison of two checkpoints should be read against the hardware each was trained on. Analysis is deterministic given a checkpoint and a seed, and every quantity reported in the article is a median or a range over the five seeds.

# 3 The control for each measurement

A contrast of this kind is interpretable only against something that would have come out differently had the instrument been responding to something else. This section states, for each row of Table 1, what that control is and what outcome from it would have refuted the reading. Each control is part of the measurement: the script that produces a deposit computes its control in the same pass, over the same rows and the same weights, so the two cannot drift apart. None of these is a pre-registration, with the one exception named at the end.

Table S4 precedes them with the arithmetic behind the summaries: every quantity the article reports as a median, a mean or a range, printed seed by seed. It is generated from the same deposits the article's Table 1 reads, by key, and its summary column is computed from the series printed beside it, so a summary and its seeds cannot disagree.

**Does the observable determine the $\sigma$-profile?** Control: the same Jacobian spectrum computed on random profiles carrying the same total area. The reading is that the deficiency belongs to the closure as a map rather than to the trained head, so it is the control's *agreement* that supports it. A control resolving materially more directions than the learned profile would refute it, because the restriction would then be something the head does. The learned profiles give a median of two directions of fifty for 99% of the information, range one to five; the random profiles give two.

**Does the observable determine the crystal term?** Control: the same Cramér–Rao bound recomputed with the melting temperature and the enthalpy of fusion supplied as independent labels. A bound already tight without them would refute the reading. It is $\pm 20\,\mathrm{kJ\,mol^{-1}}$ without and $\pm 1.7$ with.

**Does the head use the freedom?** Control: the share of the molecule's cavity surface each charge-density window holds. A residual distributed in proportion to area, that is an enrichment near one in every window, would refute the reading that the departure is directed. Over the 71 molecules of the clean pool the enrichment is 3.97 in the donor window, 2.03 in the acceptor window and 0.88 in the nonpolar core as solute, and 3.78, 1.96 and 0.89 as solvent.

**Is the output layer responsible?** Two controls, because the claim is negative and a negative needs both. The first is the smallest bin mass the same head reaches anywhere, which bounds what the normalisation makes impossible; the second is an untrained head of the same architecture, which gives the pedestal the parameterisation produces before any data. A head unable to go far below its pedestal, or an untrained head whose pedestal matched the trained one, would refute the reading. The head reaches $10^{-11}$, and the untrained pedestal is a median 0.571 against 0.02 to 0.03 trained.

**Is the reference state preserved?** Control: the same calculation with one profile supplied on both sides, which returns zero identically. A control returning anything other than zero would place the deviation in the closure rather than in the model, and would refute the reading. It returns zero exactly; the model's own two role profiles give a median $|\ln \gamma_2|$ between 0.012 and 0.044 across seeds, largest 0.80.

**What does misplaced mass cost, and where?** Control: the same per-bin sweep on random profiles of the same total area, which gives the edge-to-core ratio a smooth object produces with no chemistry in it. A real-profile ratio inside the random profiles' would refute

Table S4: Seed-by-seed values behind the quantities the article reports as a median, a mean or a range. Seeds are in ascending order, 42 to 46, except the last row, where the deposit carries three. Every series is read by key from the deposit the corresponding measurement was produced from, and the summary is computed from the series printed beside it, so a summary and its seeds cannot disagree. The summary is the statistic the article quotes for that quantity where it quotes one.

| quantity | Sec. | per seed | summary |
|---|---|---|---|
| directions carrying 99% of the information, median over solvents | 3.1 | 3 / 3 / 2 / 3 / 2 (range 1–5 over all 125 seed-solvent cells) | median 2 |
| pedestal, solute role | 3.3 | 0.021 / 0.030 / 0.023 / 0.022 / 0.021 | $0.023 \pm 0.004$ |
| pedestal, solvent role | 3.3 | 0.019 / 0.024 / 0.018 / 0.021 / 0.018 | $0.020 \pm 0.003$ |
| share of reference-zero bins below $10^{-5}$ (%), both roles | 3.3 | 34 / 35 / 41 / 42 / 44 / 46 / 47 / 47 / 49 / 51 | median 45, range 34–51 |
| median $\lvert \ln \gamma \rvert$ of a molecule in itself | 3.4 | 0.012 / 0.024 / 0.044 / 0.034 / 0.035 | median 0.034 |
| ninety-fifth percentile of the same | 3.4 | 0.09 / 0.24 / 0.34 / 0.33 / 0.25 | median 0.25 |
| edge-to-core cost ratio, training rows | 3.5 | 23.9 / 20.6 / 20.9 / 29.6 / 21.3 | median 21 |
| edge-to-core cost ratio, scoring rows | 3.5 | 17.3 / 13.3 / 14.9 / 24.8 / 13.7 | median 15 |
| random third worse than the learned profile (% of 20 draws) | 3.6 | 5 / 20 / 15 / 30 / 30 | mean 20 |
| gap recovered by a random third (%), 20× seeds | 3.6 | range −213 to +152 | +40 on average |
| gap recovered, reference into the nonzero bins (%) | 3.7 | 99 / 68 / 84 / 85 / 109 | mean +89 |
| gap recovered, exact zeros into the zero bins (%) | 3.7 | −9 / −36 / 2 / −36 / 34 | mean −9, sign varies |
| activity-coefficient AAD, learned profile | 3.8 | 1.777 / 1.662 / 1.733 / 1.748 / 1.835 | $1.751 \pm 0.063$ |
| solubility penalty on the solvent-only rows | 3.8 | 0.456 / 0.381 / 0.457 | $0.431 \pm 0.044$ |

the reading that the cost map is a property of the closure's chemistry. The real ratios are 21× and 15× on the two row sets against 1.6× random.

**Does the intermediate decompose?** Control: a randomly chosen part of the grid of the same size as the part being substituted, twenty draws per seed. If substituting a random part were never worse than substituting none, the parts would add and a per-bin attribution would be defined. A random third is worse than the fully learned profile in 20% of the $20 \times 5$ draws.

**Where does the error that reaches the closure live?** Controls: exact zeros written into the bins where the reference is zero, and a random third of the grid, twenty draws per seed. A zero-bin repair recovering a share comparable to the nonzero-bin repair would refute the reading that the consequential error is in the occupied bins. The nonzero repair recovers +89% of the gap and the zero repair −9%.

**What does supplying the reference profile do?** Control: the rows in which no profile is replaced because neither component is tabulated, which must not move. Movement there would mean the substitution was reaching rows it does not touch and would refute every number read from this contrast. Those rows move by less than $4 \times 10^{-6}$. The activity coefficients improve from $1.75 \pm 0.06$ to 0.76 and the solubility worsens by $0.43 \pm 0.04$ over 5274 rows.

**The one measurement with a rule written beforehand.** The reference-state test

was prompted by a reviewer's observation that a role-dependent profile would break $\gamma(i \text{ in } i) = 1$. A rule was committed before the quantity was computed, six minutes before the result in the repository history, and it reads as follows. The existing closure validation cannot see the effect, because it supplies one tabulated profile to both sides and so tests the closure rather than the model; under equal profiles $\gamma = 1$ holds by construction. The model is built with a role-specific branch, so one molecule receives different profiles in the two slots, and the quantity to compute is therefore $\ln \gamma_2^{\infty}$ with the solute-role profile in one argument and the solvent-role profile in the other, at 298.15 K, with the layer's default configuration. The declared control was the one-profile calculation, which must return zero exactly; the declared refutation was that control returning anything else, in which case the deviation would be the closure's. Both parts were then computed, and both came out as declared.

# 4 What the observations leave undetermined

Section 3.1 of the article states the result: two of the fifty directions in which a $\sigma$-profile can move carry 99% of the information a solubility observation supplies, and random profiles of equal area give the same count. This section gives what the article does not: the protocol, the chemical locus of the directions that are resolved, an independent measurement at infinite dilution, and the limits that travel with it.

**Protocol, and why the control agreeing is the result.** Fix the weights and the rows and perturb the $\sigma$-profile along each of the 51 grid coordinates in turn, recording the response of $\ln \gamma_2$; the information matrix is the Gram matrix of the resulting Jacobian, and the reported count is the number of eigenvalues needed to carry 99% of its trace. The control is the same construction on random profiles of equal total area. Its agreement is the finding rather than a failure of the test: the deficiency belongs to the closure as a map from a 51-bin histogram to one number, so an observation of $\ln \gamma_2$ constrains a two-dimensional shadow of the profile whatever profile is supplied, and what the head places in the remaining coordinates is not decided by fitting.

**Which directions are resolved.** The resolved direction has a chemical locus as well as a rank. The dominant eigenvector concentrates on the extreme-$|\sigma|$ bins on which the closure's hydrogen-bonding term acts, so the coordinates the observation leaves free are the nonpolar and weakly polar interior of the profile. The two measurements in the article meet here: the free coordinates are also the ones in which misplaced mass is cheapest (Section 3.5), so the network is least constrained exactly where departures matter least, and the departure that does reach the thermodynamics is a departure in magnitude rather than in location (Section 3.7).

**The same picture at infinite dilution.** A complementary measurement reaches it by another route. The aggregate activity Fisher matrix $J^{\top}J$ with $J = \partial \ln \gamma_2^{\infty}/\partial\sigma$ is rank one, and the per-molecule Fisher that governs an individual profile has participation ratio about 1.4 of 51 even at full solvent coverage.

Three limits travel with these numbers. The participation ratio is a concentration statistic and not an exact count of resolved directions. It must be read against a permuted control rather than absolutely, because finite differences decorrelate the redundant flat-profile coordinates and inflate it. And it is measured at infinite dilution, so its transfer to the finite-composition regime is not established here. None of this says that recovering the tabulated profile is impossible in principle; it says the solubility observations do not require it.

**Why the crystal term is treated differently.** The article reports the condition number of the four-parameter fit and the Cramér–Rao bound it implies. What follows for the design is the asymmetry between the two heads: the crystal term is indeterminate from solubility alone but has an external label that closes

the indeterminacy, which is why it is supervised externally, whereas the profile coordinates the closure cannot see have no such label. That asymmetry is the reason the article's question is about the profile rather than the crystal term.

# 5 A second system: a linear free-energy relation

The article's measurements are made on one pipeline with one closure, so they cannot say whether the behaviour depends on the closure's adequacy. This section reports a second system built so that adequacy varies by chemical stratum and is known from structure alone.

**Construction.** The closure is a fixed Hammett linear free-energy relation, $g = \mathrm{p}K_{a,0} - \rho\,\sigma^{\star}_{\mathrm{H}}$, over a reaction series $s$, which fixes the per-scaffold Hansch–Leo–Taft constants $(\mathrm{p}K_{a,0}, \rho)$, and a substituent constant $\sigma^{\star}_{\mathrm{H}}$. A network predicts $\sigma^{\star}_{\mathrm{H}}$ from structure and the prediction is pushed through the fixed relation, which is the same arrangement as a $\sigma$-profile pushed through COSMO-SAC. The tabulated Hansch–Leo values are the external reference for the predicted coordinate, exactly as the VT-2005 table is for the profile. The constant carries its subscript throughout so that it is not confused with the screening-charge density of the article.

Assignment is mechanical and reads structure only. Four reaction series are tried in the fixed order benzoic acid, phenol, anilinium, pyridinium, and a record enters under the first whose ionizing group matches exactly once, at a ring atom lying in exactly one six-membered aromatic ring. Ring positions are typed by shortest ring-bond distance from that atom: one bond ortho, two meta, three para. The constant $\sigma^{\star}_{\mathrm{H}}$ is the sum over substituted positions of the tabulated Hansch–Leo value for that substituent at that position, from a fixed list of nineteen; an unsubstituted parent has $\sigma^{\star}_{\mathrm{H}} = 0$.

**Why adequacy varies, and how the strata are defined.** The two classical failure modes of a Hammett relation fall on either side of the distinction the article draws. Resonance saturation is a curvature in the substituent constant that a linear relation structurally cannot represent: a defect of the map. Ortho and steric field effects are absent from the tabulated constants altogether: information the intermediate does not carry.

The strata follow from that. A record whose every substituted position is meta or para with a tabulated substituent lies where the relation is close to exact. A record carrying an ortho substituent, or one outside the tabulated list, lies where it is not; both enter $\sigma^{\star}_{\mathrm{H}}$ as zero. The rule is a function of structure and never reads a residual, so the strata cannot be chosen by outcome.

**The same distinction, dialled continuously.** Before the comparison on measured data, the two failure modes can be separated by construction. Figure S1 dials each of them on a semi-synthetic Hammett series whose true values are known in closed form, so that the cost of supplying the reference can be read against a quantity that is not estimated. Degrading the relation while the constant stays sufficient makes the cost follow the relation's own error, from $-0.03$ where the relation is exact to $+0.69$ where it is not, against the relation's own 0.81. Keeping the relation exact and starving the constant instead leaves the relation's own error at zero throughout, and the cost still rises, from $-0.03$ to $+0.34$, crossing zero between fractions of 0.15 and 0.30. Supplying the reference is therefore costly for two separable reasons, and only one of them is a defect of the map.

**Data and one uniform filter.** The source is the OPERA compilation,[S5] 7904 records. A small number are supplied as pre-ionized microstates, inconsistent with a neutral-scaffold Hammett assignment, and they degrade the well-specified stratum severely. They are removed by one rule applied uniformly to both strata, net formal charge different from zero, which never references a residual and so cannot be circular, and which keeps net-neutral substituents such as nitro. The filter removes 5 of

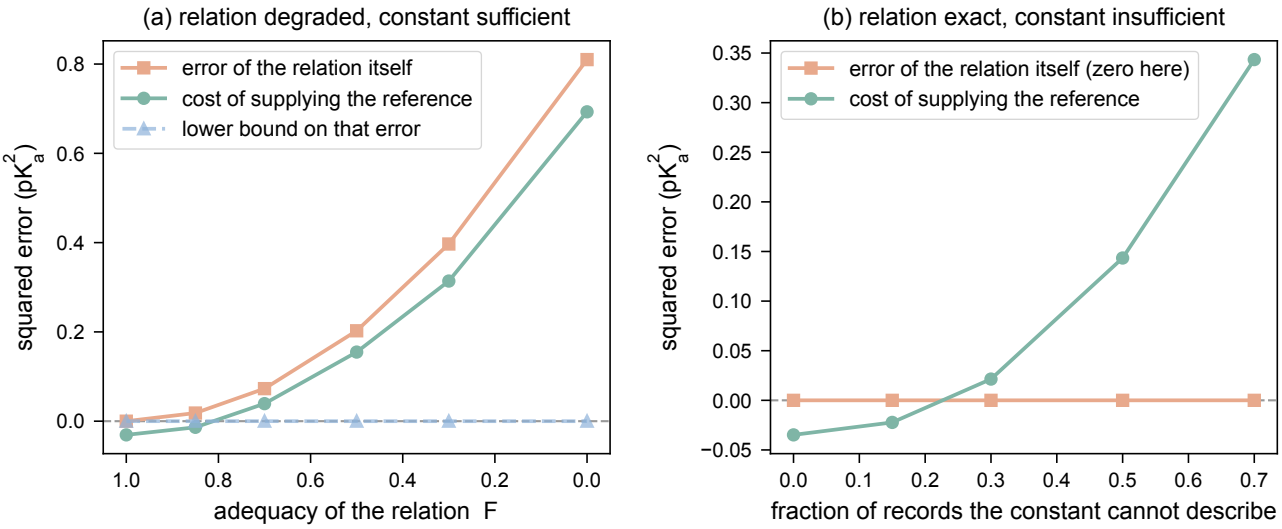



Figure S1: The two failure modes dialled directly, on a semi-synthetic Hammett series whose true values are known in closed form. The cost of supplying the reference is the squared error obtained with the tabulated constant minus that of the best free fit, so a positive value means supplying the reference makes the prediction worse. (a) The relation is degraded while the constant stays sufficient; adequacy $F$ falls from left to right. The cost tracks the relation's own error and changes sign where the relation becomes adequate. (b) The relation is kept exact while a growing fraction of records carries an effect the tabulated constant cannot describe. The relation's own error is zero throughout and the cost still rises, so a cost of supplying the reference is not by itself evidence that the relation is wrong.

163 rows in the well-specified stratum and 352 of 1198 in the other, leaving 1004 Hammett-amenable rows, 158 and 846.

**The comparison, and the result.** The physics arm, encoder to $\hat{\sigma_{\mathrm{H}}}$ to the fixed relation, is trained on $K = 20$ stratified within-scaffold splits, and the per-molecule error is paired against the deterministic reference arm, stratum by stratum. Where the relation is well specified the reference wins: learned MAE $0.48 \pm 0.09$ against $0.315 \pm 0.07$, a gap of $-0.165$, the reference better in 19 of 20 splits. Where it is not, the reference loses: learned $0.83 \pm 0.07$ against $1.62 \pm 0.08$, a gap of $+0.79$, the reference worse in 20 of 20 splits.

The sign of the effect of supplying the true intermediate therefore follows the adequacy of the relation it is supplied to, within one system, under one protocol, with the strata fixed by structure.

**Three limits.** The comparison is within distribution. The companion run on a Bemis–Murcko scaffold split is deposited beside it, and there nothing flips: both arms degrade and the learned arm degrades further, so the reference helps on both strata. The reason the main comparison is not run on a scaffold split is that such a split collapses the well-specified stratum to two ring systems. Every solubility result in the article is measured on a scaffold split, so this is the one place the protocol differs.

The resampling unit is the reaction series and only four exist, so the cluster bootstrap has 35 distinct outcomes and a finest atom of about 0.004. The interval endpoints quoted above are percentiles of that law and are no finer; read as a sign test on four units the two statements are $p = 1/16$ and $p \approx 0.31$. What the result rests on is the consistency of the sign across splits and series and the size of the gap in the misspecified stratum, $+0.79$ against a reference error of 1.62, and not on a $P$ value. The re-

fit that would settle the series-level question at four units was not run.

Finally, a competence sweep over training fractions 1.0, 0.5, 0.25 and 0.12, six seeds each, shows that the reference side of the comparison is fixed by the stratum and not by the data budget: its error is the same at every budget while the learned side moves. In the well-specified stratum the learned arm rises from 0.46 to 0.98 and never reaches the reference's 0.31, so the reference helps throughout. In the other it stays below the reference's 1.62 across the whole range, 0.85 at full data and 1.31 at 12%, so the reference hurts throughout. Only scaffold extrapolation, where the predicted constant breaks down to $2.6 \pm 2.0$, pushes it back above and reverts the sign, and that reversion is a property of the mean over six splits rather than of each: three of the six still have the reference hurting, and the negative mean is carried by two splits.

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